LESSON 1:

Introduction to Computer Graphics

What is Computer Graphics?

- A graphic is an image or visual representation of an object.
- Computer Graphics is an art of drawing pictures on computer screens.
- Creation, Manipulation, and Storage of geometric objects (modeling) and their images (rendering)
- Typical graphics system comprises of a host computer with support of fast processor, large memory, frame buffer and
  - Display devices (colour monitors),
  - Input devices (mouse, keyboard, joystick, touch screen, trackball)
  - Output devices (LCD panels, laser printers, colour printers. Plotters etc.)
  - Interfacing devices such as, video I/O, TV interface etc.

Graphics Definitions

Point: a location in space, 2D or 3D sometimes denotes one pixel

Line: straight path connecting two points

Vertex: point in 3D

Edge: line in 3D connecting two vertices

Polygon/Face/Facet: arbitrary shape formed by connected vertices, fundamental unit of 3D computer graphics

Mesh : set of connected polygons forming a surface (or object)

Graphics Definitions

Rendering : is the process of generating an image from a 2D or 3D model (or models in what collectively could be called a scene file), by means of computer programs. Also, the results of such a model can be called a rendering.

Framebuffer : a large block of memory that stores the color values for all the pixels on the screen.

What drives computer graphics?

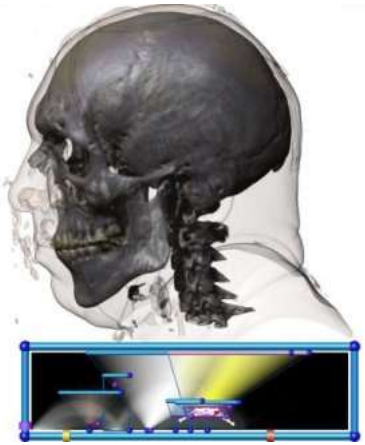
- Movie Industry
  - Leaders in quality and artistry
  - Big budgets and tight schedules
  - Defines our expectations

- Game Industry
  - The newest driving force in CG due to volume and Profit
  - Focus on interactivity
  - Cost effective solutions

What drives computer graphics?

- Medical Imaging and Scientific Visualization
  - Tools for teaching and diagnosis
    - No cheating or tricks allowed
  - New data representations and modalities
  - Drive issues of precision and correctness
  - Focus on presentation and interpretation of data
  - Construction of models from acquired data

Nanomanipulator, UNC



Joe Kniss, Utah

Gordon Kindelman, Utah

What drives computer graphics?

- Computer Aided Design
  - Drives the high end of the hardware market
  - Integration of computing and display resources
  - Reduced design cycles results in faster systems in a shorter timeframe

- Education and Training
  - Models of physical, financial, social systems
  - Comprehension of complex systems

- Computer Art
  - Fine and commercial art
  - Performance Art
  - Aesthetic Computing

Historical Milestones

- 1960's:
  - Early theoretical development, mainly limited to research and military
  - 1962: Sketchpad (Ivan Sutherland)
- 1970's:
  - 'Traditional' graphics pipeline developed
  - Driven by money from military simulation and automotive design industries
- 1980's:
  - Many important core algorithms developed
  - Visual quality improved driven by demands from entertainment (movie) industry
  - 1985: Rendering Equation (James Kajiya)
- 1990's:
  - Advanced algorithms developed as graphics theory matured
  - Broader focus on animation, data acquisition, NPR, physics…
  - 1995: Photon Mapping (Henrik Jensen)
- 2000's:
  - Photoreal rendering evolves to the point of being able to render convincing images of arbitrarily complex scenes on consumer hardware
  - Merging of computer graphics and computer vision
  - Cheap graphics hardware with vast capabilities, driven largely by video game industry

Image Processing

- Some computer graphics operations involve manipulating 2D images (bitmaps)
- *Image processing* applies directly to the pixel grid and includes operations such as color correction, scaling, blurring, sharpening, etc.
- Common example include digital photo processing and digital ‘painting’ programs (Adobe Photoshop…)

Image Synthesis

- *Image synthesis* or *image generation* refers more to the construction of images from scratch, rather than processing of existing images

- Synthesis of a 2D image from a 3D scene description is more commonly called *rendering*

Photoreal Rendering

- *Photoreal rendering* refers to rendering a 3D scene in a realistic way

- Modern photoreal rendering algorithms are essentially a physically based simulation of light propagation and scattering throughout a 3D environment

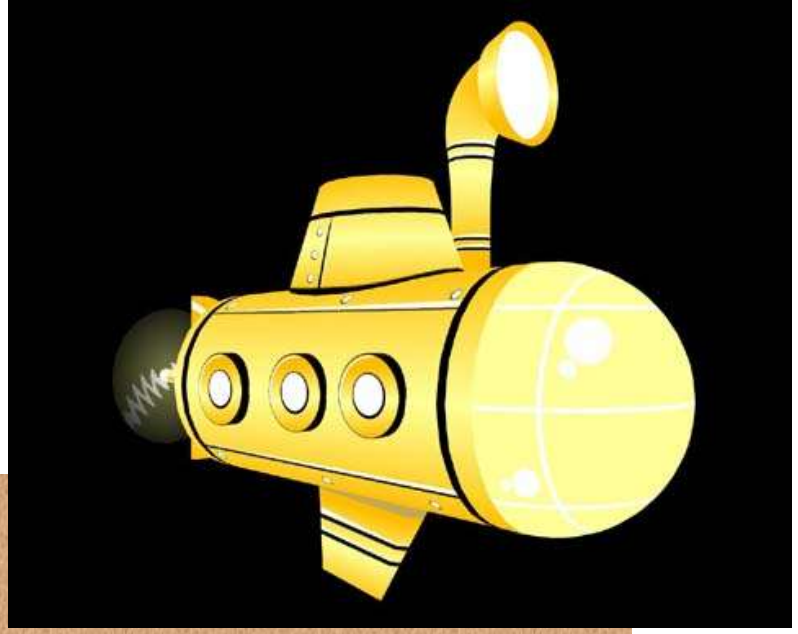
- This means that there is a ‘correct’ image that should be generated, given an input data set. This allows the subject of photoreal rendering to have a strong theoretical basis (namely, the science of optics)

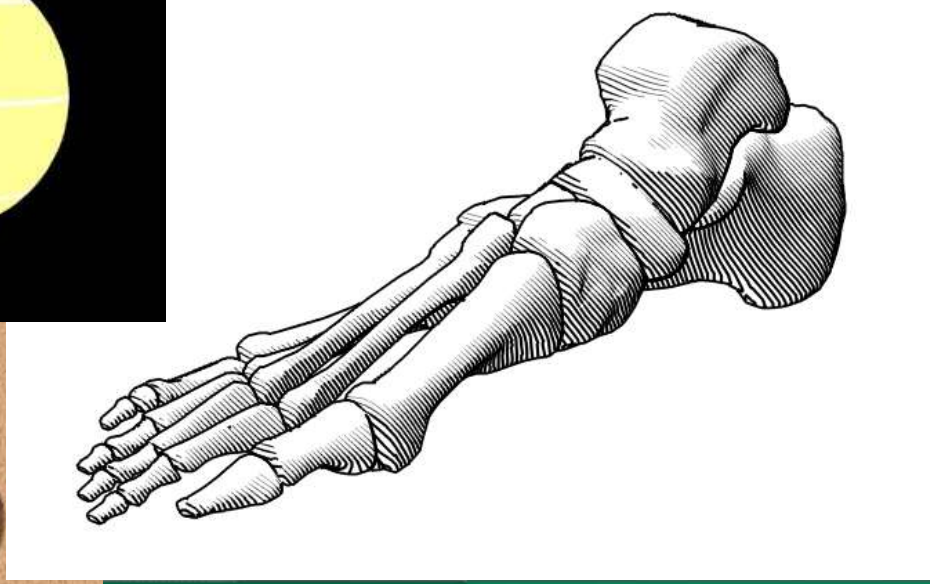
- Most modern photoreal rendering algorithms are based on the classic *ray tracing* algorithm, that traces the path of individual light rays starting from the eye and working backwards to the light sources.

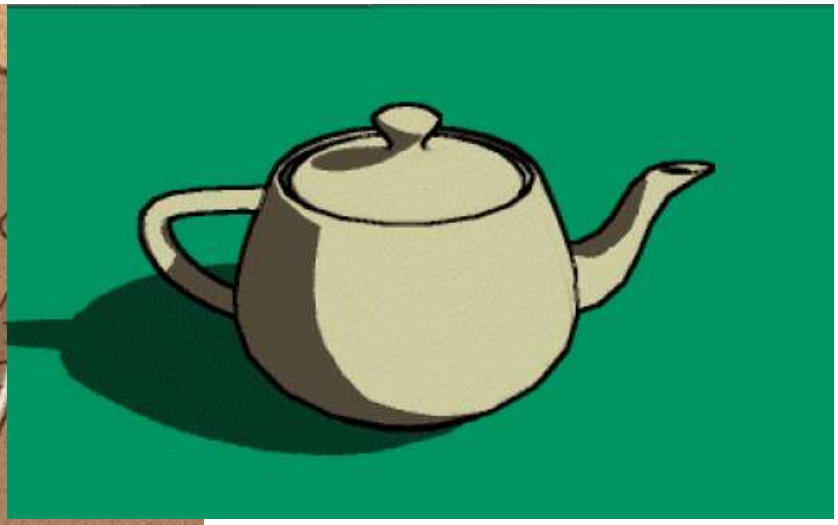
Non-Photoreal Rendering

- *Non-photoreal rendering* (NPR) refers to rendering images in other ways…
- Sometimes, this is done to achieve aesthetic goals such as artificial water colors, pencil sketches, paint brushstrokes…
- Other times, the goal is to maximize the communication of visual information, as in scientific and medical visualization

NPR







Computer Vision

- Computer vision is sometimes considered as a separate discipline from computer graphics, although they share many things in common

- A central goal in computer vision is to take a set of 2D images (usually from a video or set of photos) and infer from that a 3D description of what is being viewed

- This is a very different process than rendering, and is more of a form of artificial intelligence

Animation

- An animation is just a sequence of individual images
- Basically, the subject of computer animation focuses on how things change over time. Usually, this refers to motion, but can also refer to other properties changing over time.
- Physical simulation is a very powerful tool in computer animation and can be used to generate believable animations of rigid objects, deformable objects, gasses, liquids, fracture, particle effects, and even explosions and fire
- Computer animation also includes a large number of techniques specifically developed to manipulate virtual characters

Modeling

- *Modeling* refers to the techniques involved with creating, scanning, editing, and manipulating 3D geometric data
- It is done by a human user with an interactive editing program
- More complex objects, such as trees, can be constructed with automatic *procedural modeling* algorithms
- 3D models are also often acquired from real world objects using laser scanning or computer vision techniques
- Modeling also includes the use of curved surfaces and other higher order primitives, which are often converted into triangles using various *tessellation* algorithms
- It also includes mesh reconstruction for surface simplification
- Modeling makes heavy use of *computational geometry*

Display Technology

- Vector display (cathode ray tube)

- Raster displays
  - CRT (cathode ray tube)
  - LCD (liquid crystal display)
  - TFT (thin film transistor)
  - OLED (organic light emitting diode)
  - Light valve
  - Plasma
  - HDR (high dynamic range: TFT / white LED hybrid)

- Film
- Print

Raster Graphics

- Modern graphics displays are *raster* based
- They display a grid of pixels, where each pixel color can be set independently
- Individual pixels are usually formed from smaller red, green, and blue subpixels. If you look very closely at a TV screen or computer monitor, you will notice the pattern of subpixels
- Older style *vector* displays didn't display a grid of pixels, but instead drew lines directly with an electron beam
- Raster graphics are also sometimes called *bitmapped graphics*

Interlacing

- Older video formats (NTSC, PAL) and some HD formats (1080i) use a technique called *interlacing*
- With this technique, the image is actually displayed twice, once showing the odd *scanlines*, and once showing the even scanlines (slightly offset)
- This is a trick for achieving higher vertical resolution at the expense of frame rate (cuts effective frame rate in half)
- The two different displayed images are called *fields*
- NTSC video, for example, is 720 x 480 at 30 *frames* per second, but is really 720 x 240 at 60 *fields* per second
- Interlacing is an important issue to consider when working with video, especially in animation as in TV effects and video games
- Computer monitors are generally not interlaced

Framebuffer

- The *framebuffer* refers to the memory dedicated to storing the image
- It would generally be a 2D array of pixels, where each pixel stores a color (Note: pixel = picture element)
- Color is typically stored as a 24 bit RGB value. This offers 8 bits (256 levels) for red, green, and blue, for a total of 16,777,216 different colors
- Very often, additional data is stored per pixel such as depth (z), or other info
- A framebuffer can just be a block of main memory, but many graphics systems have dedicated framebuffer memory with a direct connection to video scan-out hardware and other special features

Primitives

- Complex *scenes* are usually built up from simpler *objects*
- Objects are built from individual *primitives*
- The most common and general purpose 3D primitive is the triangle
- Points and lines are also useful primitives

- Four basic output primitives (or elements) for drawing pictures:
  - Polyline
  - Filled POLYGONS (Regions)
  - Ellipse (Arc)
  - Text
  - Raster IMAGE

Traditional Graphics Pipeline

- In the *traditional graphics pipeline*, each primitive is processed through the following steps:
  1. Transformation
  2. Lighting
  3. Clipping
  4. Scan conversion
  5. Pixel processing

1. Transformation

- The transformation process refers to the linear transformation from 3D space to a 2D viewing space
- Each vertex position must be transformed from it's defining *object space* to the *device coordinates* (pixel space)
- This often involves a combination of rotations, translations, scales, and perspective transformations

2. Lighting

- Lighting operations are applied to each vertex to compute its color
- In more advanced rendering, lighting operations are computed per pixel, rather than per vertex
- A variety of light types can be defined such as point lights, directional lights, spot lights, etc.
- More advanced lighting operations can account for shadows, reflections, translucency, and a wide variety of optical effects

3. Clipping

- Some triangles will be completely visible on the screen, while others may be completely out of view
- Some may intersect the side of the screen and require special handling
- The camera's viewable space forms a volume called the *view volume*. Triangles that intersect the boundary of the view volume must be *clipped.*
- The related process of *culling* refers to the determination of which primitives are completely invisible
- The output of the clipping/culling process is a set of visible triangles that lie within the dimensions of the display device

4. Scan Conversion

- The *scan conversion* (or *rasterization*) process takes 2D triangles as input and outputs the exact pixels covered by the triangle

- Per-vertex data, such as color, is *interpolated* across the triangle, so each pixel may have a unique color

5. Pixel Processing

- The output of the scan conversion process is a bunch of individual xy pixels, plus additional data per pixel such as interpolated depth (z), color, or other information
- The pixel processing stage includes the operations that take place per pixel to compute the final color that gets rendered into the *framebuffer*
- Usually, the *zbuffer* technique is used to make sure that a pixel is rendered only if it is not blocked by an existing surface
- Other processing, such as texturing and transparency operations happen per pixel
- In some systems, the entire lighting process is computed per pixel, instead of per vertex

Scene Rendering

- With the traditional *zbuffered* graphics pipeline, triangles can be rendered in any order without affecting the final image
- Often, complex effects such as transparency, actually do depend on the rendering order, and so may require additional care
- Still, it makes a nice basic approach, and it's the approach taken by OpenGL and built into many modern hardware graphics boards
- There are more advanced rendering algorithms (scan line, ray tracing, etc.) that don't render triangles one at a time, and require the entire scene to be processed.